Paper 6: Conservation Laws, Records, and Time-like Structure

The $\Sigma\lambda^2$ Non-conservation Theorem, the Freezing Theorem, the B_4 Gauge, the 1+3 Polar Decomposition, the Record Theorem, and the Null Structure

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Version: v0.4 (PDF glyph typesetting fix only; md content and results unchanged — subscripts/superscripts, floor/ceil, calligraphic N, QED square, double arrows, ballot marks mapped to LaTeX to remove tofu) (content and results unchanged from v0.2; nomenclature unified to the displacement-record definition of Paper 0.5 — ledger/register sorted into displacement record / conserved quantity / invariant / accounting. The λ side = displacement record, and the direction (arrow of time) is attributed to the asymmetric one bit rather than to monotonicity, consistent with Paper 0.5.)
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Abstract

This paper establishes the kinematic core of the reciprocal dual model (Papers 1–5). The assumptions are only three — (1) the reciprocal duality $\nu\lambda = 1$, (2) the zero point 1/2, and (3) one asymmetric bit, namely that the additive conservation law rides on only one of the two dual sides — and this paper organizes, as theorems, how much structure follows from this inventory.

First, the **freezing theorem**: under conservation of $\sum \nu^2$, every nontrivial split strictly increases $\sum \lambda^2$ (a one-line proof by Cauchy–Schwarz). As a corollary, if both sums of squares were conserved simultaneously, no nontrivial split would exist, and no cascade or time-like structure would arise. **The existence of the asymmetric one bit is itself the existence condition for time-like structure.**

Second, the **gauge structure**: the exact symmetry group of the counting condition is the hyperoctahedral group B_4 (order 384), and within each energy shell the distribution over axes, the signs, and the orientations are pure gauge. The additive conservation laws close the accounting with just two entries, $\sum \nu^2$ and the Z_2 parity. As a by-product, the fine structure of shells that are degenerate in energy and split by shape (137 states at $R = 3 \to 7$ gauge-invariant classes) is laid bare.

Third, the 1+3 **polar decomposition**: it is not that one of the four axes becomes time. In the polar decomposition $\mathbb{R}^4 \cong \mathbb{R}_+ \times S^3$ of the 4-dimensional frequency space, the radial direction, on which the manifestation of the asymmetric bit concentrates, appears as time-like, while the angular directions untouched by the duality (the B_4 gauge) appear as space-like. The 1+3 role differentiation is obtained without introducing the imaginary unit or the Minkowski negative sign. We also show that there is no way to decide from the inside whether one is at the apex of the hierarchy (hierarchy relativity).

Fourth, the **record theorem and the null structure**: measurement is recording, and recording is accumulation.

Since only a non-conserved monotone quantity (the λ side) can accumulate, every displacement record remains as a configuration on the λ side, and ν can never be a direct read-out target. The integer lattice is reinterpreted not as an assumption but as the **resolution structure of unit records** (a downgrade in the assumption inventory). Furthermore, in the log-conjugate plane $(q, p) = (\log \lambda, \log \nu)$ the constraint $\nu\lambda = 1$ is a **null line**, the freedom of distributing the conjugate widths is the $SO(1, 1)$ boost (squeeze), and uncertainty is built in as the boost-invariant minimal area. The negative sign is not an axiom but a hyperbola — though the aggregation of the four pairs of $(1, 1)$ structures into a single $(1, 3)$ signature remains as an explicit construction task.

This paper does not claim to have derived physical time, space, or measurement. What it does claim is that, from the inventory of three assumptions, the role differentiation into time-like, space-like, record, uncertainty, and null structure follows as theorems.

Keywords

conservation laws, arrow of time, records, relational time, hyperoctahedral group, gauge redundancy, polar decomposition, null structure, squeezing, uncertainty, hierarchy relativity

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1. Introduction

Through Paper 5 [3], the mathematical foundation of the reciprocal dual model (Paper 1 [1], Paper 4 [2]) — the dictionary, the coherence conditions, and the classification of R^2 — was established. The question of this paper is the following.

> In this model, from which assumptions, and to what extent, does the structure in which "time appears to flow, space appears to extend, and records appear to remain" follow as theorems?

The standpoint that does not regard time as a fundamental quantity but constructs it from relations or records has a long lineage: the "frozen formalism" in the canonical formulation of quantum gravity (DeWitt [5]), time as conditional probability (Page & Wootters [6]), relational quantum mechanics (Rovelli [7]), the elimination of time from dynamics (Barbour [8]), and quantum theory on a deterministic substrate ('t Hooft [9]). This paper does not extend these theories; its distinctive feature is that it implements the problem awareness that "time is not fundamental" as a family of theorems **exhaustively verifiable within a finite lattice model**.

Organization of this paper: §3 treats the freezing theorem, §4 the B_4 gauge and the fine structure, §5 the 1+3 polar decomposition and hierarchy relativity, §6 the record theorem, and §7 the null structure and uncertainty.

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2. Inventory of Assumptions

The assumptions used in this paper are the following three (with configuration reading and the record principle used jointly as operating principles).

1. **Reciprocal duality** $\nu_n\lambda_n = 1$.
2. **Zero point** $1/2$ (structurally identified as the zero-point quantum in Paper 5).

3. **Asymmetric one bit**: the additive conservation law $\sum \nu^2 = S$ rides on one side only (the ν side).

An important remark: the labels ν/λ in Assumption 3 are not proper names but **markers of roles**. The assumption reduces to the naming convention that the observer calls the side that appears conserved the "frequency," and the only remaining physical content is the single bit that "one asymmetry exists." The mirror model (calling $\sum \lambda^2$ the conserved quantity) is merely a relabeling of the same structure.

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3. The Freezing Theorem: Asymmetry Is the Existence Condition for Time-like Structure

3.1 The $\Sigma\lambda^2$ non-conservation theorem

Theorem: Under conservation of $\sum \nu^2$ ($s = \sum_a s_a$, $s_a \geq 1$), every nontrivial split (into $n \geq 2$ parts) strictly increases $\sum \lambda^2 = \sum 1/s_a$.

Proof: By Cauchy–Schwarz, $(\sum s_a)(\sum 1/s_a) \geq n^2$, that is, $\sum 1/s_a \geq n^2/s > 1/s$ (for $n \geq 2$). ■

Numerical confirmation over all 68 coherent partitions of odd $s \leq 15$ likewise shows no exception.

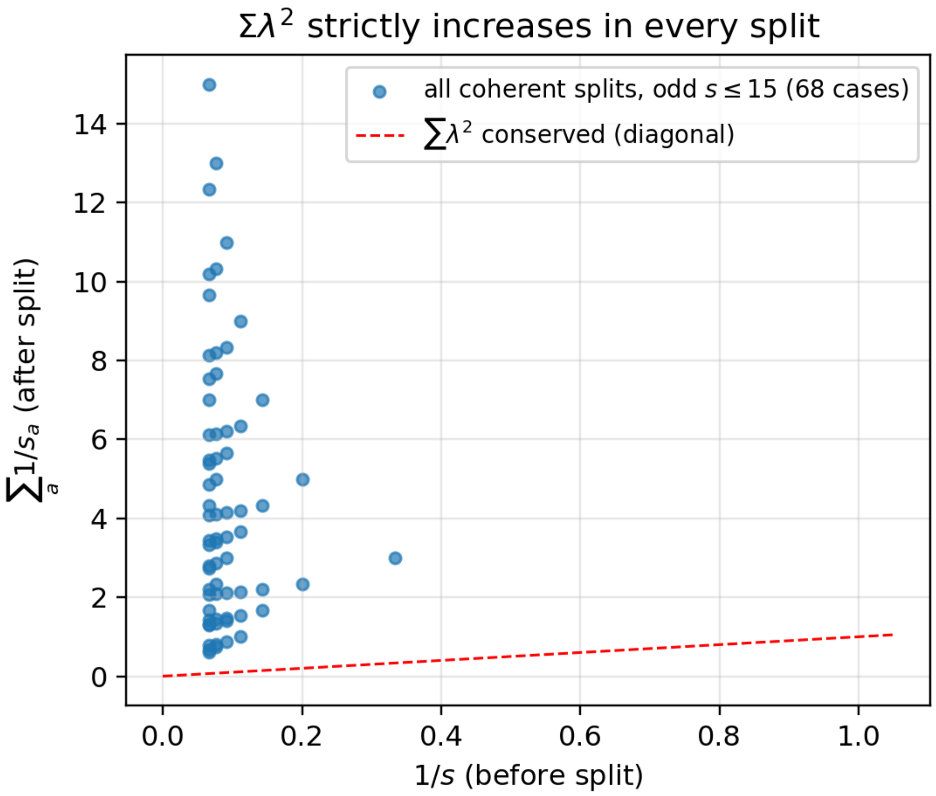

図 1　Figure 1. Strict monotonicity of the lambda side (displacement record) under every coherent split

Figure 1. The $\Sigma\lambda^2$ monotone increase theorem: all 68 coherent partitions of odd $s \leq 15$ lie above the diagonal (exact computation).

3.2 The freezing theorem

Corollary (freezing theorem): If both $\sum \nu^2$ and $\sum \lambda^2$ were conservation laws, no nontrivial split would exist at all, and no cascade, time-like structure, or expansion-like structure would arise (the system freezes).

That is, the fact that $\sum \lambda^2$ is **not** conserved — the existence of the asymmetric one bit — is itself the condition under which time-like structure can exist. The λ side (the displacement record) is the only monotone quantity in this model, and its monotone increase provides time-like structure (the possibility condition for ordering); the direction — the "arrow of time" itself — does not follow from monotonicity alone but is attributed to the asymmetric one bit of which dual side the conservation law rides on (consistent with Paper 0.5). Note that whereas the classical correspondence between symmetries and conservation laws (Noether [4]) determines *what* is conserved, the asymmetric one bit here is an independent choice of *which dual side the conservation law rides on.*

3.3 Invariance under dual relabeling

In log space the duality is the point symmetry $p = -q$, and the kinematic content of the splitting cascade (ratios, exponents, dimensionless quantities) is invariant under the relabeling $\nu \leftrightarrow \lambda$. The only element that does not disappear under the relabeling is the single bit of "which side is additive," and this is the precise content of Assumption 3 in §2.

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4. The B_4 Gauge Structure and the Fine Structure of Shells

4.1 Symmetry group and orbit decomposition

The exact symmetry group of the counting condition $\sum(|k_i| + \frac{1}{2})^2 \leq R^2$ is the hyperoctahedral group B_4 (signed permutations of the coordinates, of order $2^4 \cdot 4! = 384$; for the general theory see Humphreys [10]). The B_4 orbits of shell m are labeled by the shape (the multiset of the $|k_i|$), and the orbit size is given by

$$|\mathcal{O}(\text{shape})| = \frac{4!}{\prod_v m_v!} \times 2^{\#\{i:|k_i|>0\}}$$

(verified to agree with the shell counts for all shells with $m \leq 25$).

4.2 Gauge nature and closure of the accounting

Within each orbit, B_4 acts transitively. Therefore the distribution over axes, the signs, and which axes are excited are **pure gauge** — they are carried into one another by lattice rotations — and require no fixing by conservation laws.

> **Closure of the accounting**: The non-gauge content of a state consists of the two items (m, shape). The additive conservation laws are exactly two — $\sum \nu^2$ and the Z_2 parity (Paper 5) — and the accounting of degrees of freedom closes.

4.3 Fine structure of shells

The structure of shells degenerate in energy (m) and split by shape first appears at $m = 7$ ($40 = 32 + 8$). The number of states after gauge reduction is 7 at $R = 3$ and 27 at $R = 5$, dramatically smaller than the nominal capacities (137, 1545). These "shape multiplets" will play the role of physical quantum numbers in subsequent papers.

Figure 2. B_4 shell fine structure (exhaustive for $m \leq 25$): shells degenerate in energy split into shape orbits (first splitting at $m = 7$).

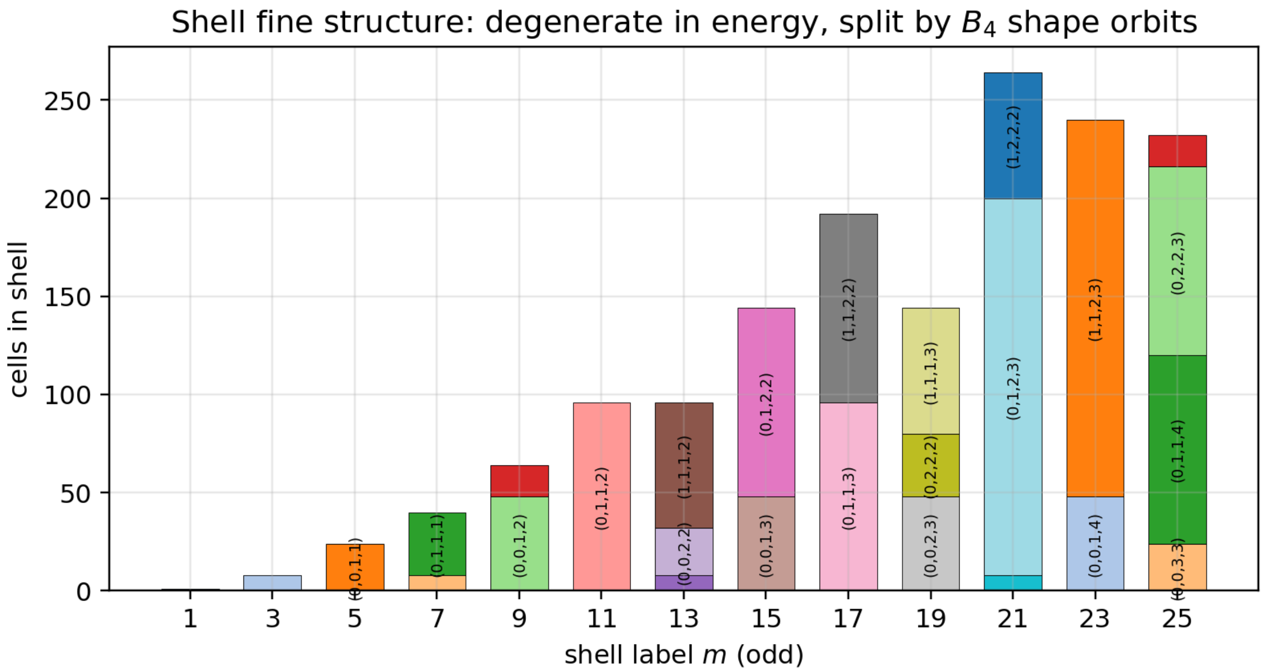

図 2　Figure 2. B4 shell fine structure

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5. The 1+3 Polar Decomposition and Hierarchy Relativity

5.1 Radial = time-like, angular = space-like

In the polar decomposition $\mathbb{R}^4_\nu \cong \mathbb{R}_+ \times S^3$ of the 4-dimensional frequency space:

- **Radial direction**: the internal clock $\tau = 1/R'$ is a function of the radius alone, and the motion of the splitting cascade is purely radial. Within a hierarchy level, the manifestation of the asymmetric bit concentrates on the radial quantity R'. Hence time-like.
- **Angular directions**: the shell degeneracy is a discretized S^3, and the angular degrees of freedom are the B_4 gauge (§4) — neither energy nor clock. The log-space duality $p = -q$ preserves the structure of the radial lines and does not touch the angles. Hence space-like.

  The B_4 symmetry of the four axes is never broken. What is "broken" is not an axis but a different way of slicing: radial versus angular. The role of the complex phase is taken over by the cos/sin doubling of the real standing-wave dictionary, and the role of the Minkowski negative sign by the null structure of §7 — **neither the imaginary unit nor the negative sign is introduced**.

5.2 Intrinsic distortion

The zero point 1/2 strictly breaks metric flatness: it produces an effective boundary (the Weyl term, Paper 5 §4) on a torus that has no boundary, and necessarily creates an unfilled layer around every state. A perfectly flat state space is the zero-point-free limit, and no state can exist there. **To exist and to be distorted are the same condition.**

5.3 Hierarchy relativity

Viewed from one hierarchy level above, the whole system is indistinguishable from an ordinary cell with the same spectrum (odd numbers, Paper 5), the same zero point, and a single representative label. Therefore:

> There is no way to determine from the inside that one is at the apex of the hierarchy. The distinction between "whole" and "part" is a hierarchy-indexical term, not a structural attribute.

The only absolute scale is the zero point $1/2$, the minimal cell, and this closes self-referentially in the form that the lowest level of the hierarchy defines the unit. All observables of an internal observer are ratios, and the structure is completely relational.

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6. The Record Theorem

6.1 Theorem

Measurement is recording, and recording is accumulation. Only a non-conserved, monotone quantity (the displacement record) can accumulate. By the theorem of §3:

- $\sum \nu^2$: invariant under all splits (exchange only is possible; accumulation is impossible).
- $\sum \lambda^2$: the only monotone quantity (the displacement record), strictly increasing under all splits.

> **Corollary (record theorem)**: Every measurement record can only remain as a configuration on the λ side. ν can be estimated only via exchange and counting, and can never be a direct read-out target.

The reason for the von Neumann-type observation [11] that every actual measurement terminates in a position reading (the location of a pointer, a photographic plate, a detector) appears in this model as a theorem.

6.2 Integer lattice = resolution structure (downgrading an assumption)

The overlap for discriminating a frequency difference $\Delta\nu$ within an observation window T is $|\mathrm{sinc}(\pi\Delta\nu T)|$, and the resolution is $\delta\nu = 1/T$. ν cannot be read in a snapshot; the price is duration. Since, with the unit record window $T = 1$, discrimination becomes exact precisely for **integer differences**:

> The integer ν lattice is reinterpreted not as an assumption about ν, but as **the structure of what a unit record can resolve**.

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7. The Null Structure: $\nu\lambda = 1$ Is the Light Cone of the Log-Conjugate Plane

7.1 Null line and boosts

Taking $(q, p) = (\log\lambda, \log\nu)$ for each conjugate pair, $\nu\lambda = 1 \iff u \equiv q + p = 0$. **The constraint is a null line.** The cascade — the time-like motion — is translation along this null line (the $v = q - p$ direction), while the direction off the cone (u) is frozen at the zero point and is paid only as uncertainty.

The freedom of distributing the conjugate widths, $S_\eta : (\delta_\nu, \delta_\lambda) \to (e^{-\eta}\delta_\nu, e^{\eta}\delta_\lambda)$, strictly preserves the area $\delta_\nu\delta_\lambda$, and this is precisely the $SO(1,1)$ hyperbolic rotation — the **boost** — of the (u, v) plane.

7.2 Uncertainty = boost-invariant minimal area

The boost invariant is the area, and its minimal value is the zero point $1/2$.

> The structure corresponding to the uncertainty principle is built in as the **boost-invariant minimal cell** of the $(1,1)$ geometry. The negative sign is not an axiom but a hyperbola.

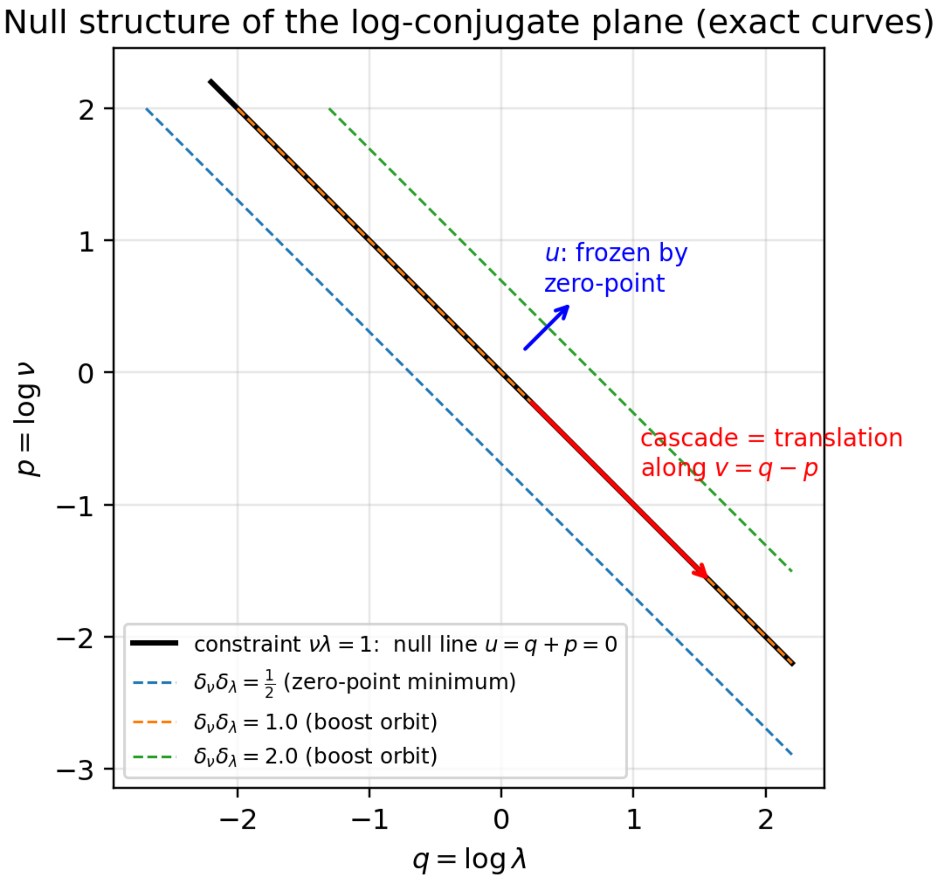

図 3　Figure 3. Null structure of the log-conjugate plane (exact curves)

Figure 3. Null structure of the log-conjugate plane (exact curves): the constraint $\nu\lambda = 1$ is a null line, the distribution freedom consists of boost orbits (hyperbolas), and the minimal area is the zero point $1/2$.

7.3 What remains explicitly unconstructed

The $(1,1)$ null/boost structure derived in §7 exists **per conjugate pair** ($(1,1)^{\times 4}$). Assembling it into a single $(1,3)$ Minkowski signature requires a specification that aggregates a single time direction out of the four pairs; the natural candidate is the radial direction (§5), but the explicit construction of the aggregation map is outside the scope of this paper. We note in passing that, even in special relativity, the $(1,3)$ signature and the dimensionality themselves are empirical inputs, not derived results.

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8. Scope of Claims of This Paper

What this paper claims: (1) the $\Sigma\lambda^2$ non-conservation theorem and the freezing theorem (with proof); (2) the asymmetric one bit = the existence condition for time-like structure; (3) the B_4 gauge nature, the closure of the accounting, and the fine structure of shells (exhaustively verified); (4) the 1+3 polar decomposition (radial = time-like, angular = space-like) and hierarchy relativity; (5) the record theorem and the reinterpretation of the integer lattice as a resolution structure; (6) the null structure, boosts, and the invariant area (per pair).

What this paper does not claim: (1) the derivation of physical time, space, or measurement; (2) the construction of the $(1, 3)$ Minkowski metric; (3) a continuous action of the Lorentz group; (4) any extension or modification of the theories of relational time [5–9]. The terms "time-like" and "space-like" are restricted to describing the role differentiation within the model.

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9. Conclusion

From the three assumptions ($\nu\lambda = 1$, the zero point 1/2, and the asymmetric one bit), the following follow as theorems with no additional assumptions: (i) the existence condition for time-like structure (the freezing theorem); (ii) the gauge nature of the space-like degrees of freedom (B_4); (iii) the 1+3 role differentiation (the polar decomposition); (iv) the asymmetry of records (the record theorem); and (v) the geometry of uncertainty (the null structure and the invariant area). The direction of time, the accumulation of records, and the asymmetry of metrizability are all different faces of one and the same single bit.

With the integer lattice downgraded from an assumption to a resolution structure, the assumption inventory of the model has become lighter still. The subsequent papers build, on top of this kinematic core, the configuration statistics (Paper 7), the two accountings (Paper 8), and the stability of composites (Paper 9).

Appendix: Essentials of the Reproduction Procedure

The numerical confirmation of $\Sigma\lambda^2$ non-conservation covers all 68 coherent partitions of odd $s \leq 15$; the B_4 orbit decomposition is checked by classifying all cells with $|k_i| \leq 6$ and comparing against the orbit-size formula ($m \leq 25$); the sinc discrimination table is evaluated on the grid $\Delta\nu, T \in \{0.25, \dots, 4\}$; squeeze invariance is confirmed to machine precision for 5 random pairs. Scripts and research notes are provided in the public repository (https://github.com/WurabeSeiji/ai-chat-logs-open).

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